

Chapter 14: Reporting & Analytics

PropMetrik provides a comprehensive reporting suite that spans every module on the platform. From project-level compliance reports to cross-portfolio valuation analytics and sensitivity modeling, this chapter covers how to generate, interpret, and export reports across the platform.

14.1 Project Reports

Generate and manage construction project reports at **Dashboard > Projects > [Project] > Reports**, or view aggregated reports at **Dashboard > Analytics**.

Available Project Report Types

Report	Description
Progress Report	Summarizes construction progress by phase, including percentage complete, milestones hit, and schedule variance
Cost Report	Breaks down actual vs. budgeted costs by cost code, with variance analysis and forecasting
Safety Report	Compiles safety inspection results, incident counts, and compliance scores
Quality Report	Aggregates checklist inspection scores, punch list status, and deficiency trends
Meeting Minutes	Formatted record of project meetings with action items and attendee lists

Change Order Summary	Lists all approved, pending, and rejected change orders with cost impact
Closeout Report	Final project summary covering schedule performance, budget adherence, and lessons learned

Generating a Project Report

1. Navigate to the project and click the **Reports** tab.
2. Select the **report type** from the template gallery.
3. Configure report parameters:
 - **Date range** -- The reporting period (e.g., this month, this quarter, project-to-date).
 - **Sections** -- Toggle which sections to include or exclude.
 - **Format** -- Choose PDF, Excel, or on-screen preview.
4. Click **Generate Report**.
5. The report appears in the report list with a download link. Click to view or download.

Scheduling Automated Reports

1. On the Reports tab, click **Schedule**.
2. Choose the **report type** and **frequency** (daily, weekly, biweekly, monthly).
3. Set the **delivery method**:
 - **Email** -- Enter recipient email addresses.
 - **In-app** -- Report appears in the notifications center.
 - **WhatsApp** -- Summary sent via the WhatsApp bot to specified numbers.
4. Click **Save Schedule**. Reports generate automatically at the configured interval.

E-Signing Reports

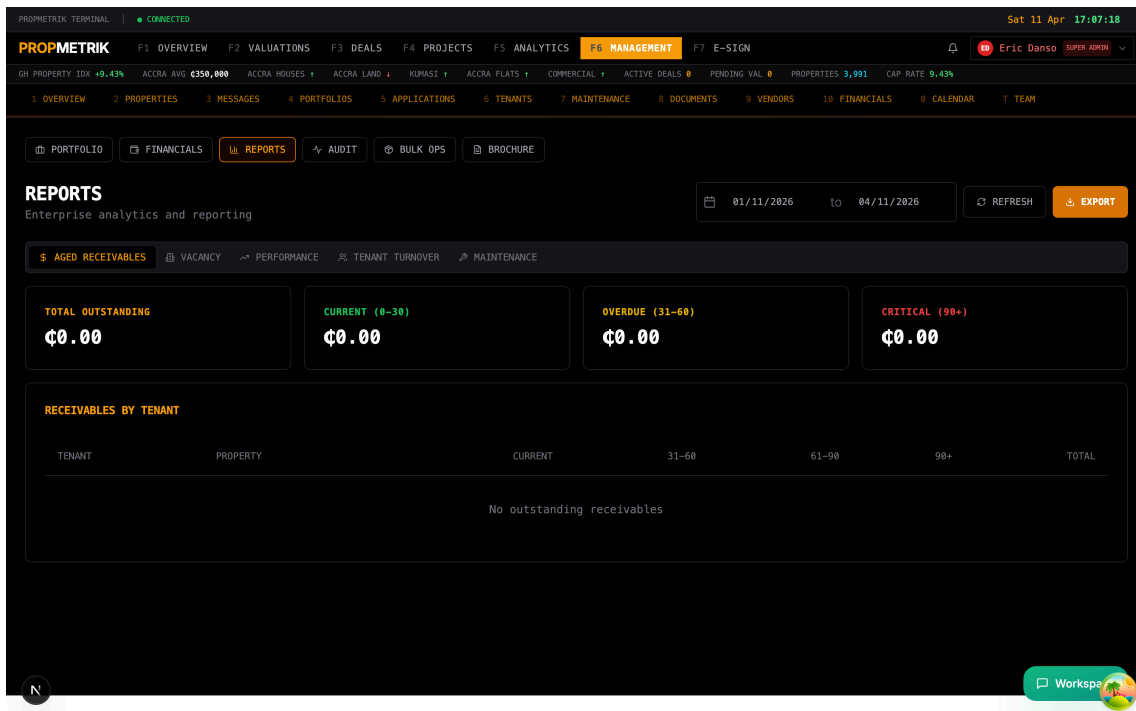
Project reports that require stakeholder approval can be routed through the e-sign workflow:

1. After generating a report, click **Send for Signature**.
2. Select recipients (project owner, client, compliance officer).
3. The report is wrapped in an e-sign envelope and sent via email with a magic link.
4. Track signing status directly from the Reports tab.

Tip: Schedule weekly progress reports to auto-generate every Friday. This creates a consistent paper trail for stakeholders and reduces manual reporting effort.

14.2 PM Reports (Project Management Reports)

Access project management analytics and compliance reports at **Dashboard > Analytics > PM Reports**.



Compliance Report Dashboard

The PM reports page provides a compliance-focused view across all active projects:

- **Overall Compliance Score** -- A weighted average of safety, quality, documentation, and schedule adherence scores across all projects.
- **Project Health Matrix** -- A grid showing each project's status across multiple compliance dimensions (red/amber/green).
- **Overdue Items** -- Count of overdue RFIs, submittals, inspections, and action items.
- **Non-Conformance Tracker** -- Lists open non-conformance reports (NCRs) with severity ratings.

Generating Compliance Reports

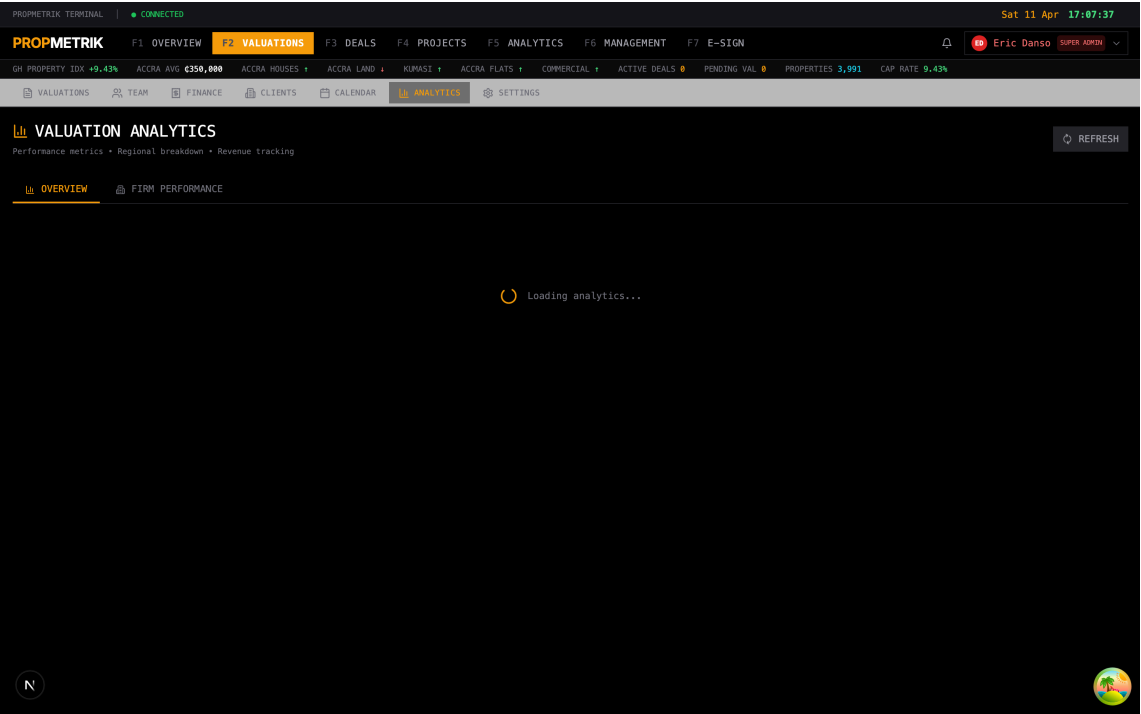
1. Select the **report scope** (single project, portfolio, or organization-wide).
2. Choose the **compliance framework** (internal standards, building code requirements, or client-specific requirements).
3. Set the **reporting period**.
4. Click **Generate**. The report includes:
 - Executive summary with overall compliance score
 - Section-by-section breakdown (safety, quality, documentation, schedule)
 - Charts showing compliance trends over time
 - Detailed findings with photos and inspector notes
 - Recommended corrective actions

Exporting PM Reports

- **PDF** -- Formatted report suitable for client distribution and regulatory submission.
- **Excel** -- Raw data export with all metrics and underlying data points.
- **API** -- Programmatic access to report data via the PropMetrik REST API.

14.3 Valuation Analytics

Access valuation performance analytics at **Dashboard > Analytics > Valuations**.



Key Metrics

The valuation analytics dashboard tracks:

- **Total Valuations** -- Count of completed valuations with period-over-period comparison.
- **Average Turnaround** -- Mean time from instruction to report delivery.
- **Revenue** -- Total valuation fee revenue by period.
- **Pipeline Value** -- Aggregate market value of properties currently under valuation.

Production Charts

- **Valuations by Month** -- Bar chart showing monthly production volume.
- **Value Distribution** -- Histogram of property values across completed valuations.
- **By Property Type** -- Breakdown by residential, commercial, industrial, and mixed-use.
- **By Location** -- Geographic distribution of valuations across regions and districts.
- **By Method** -- Split between income approach, market comparison, and cost approach valuations.

Valuer Performance

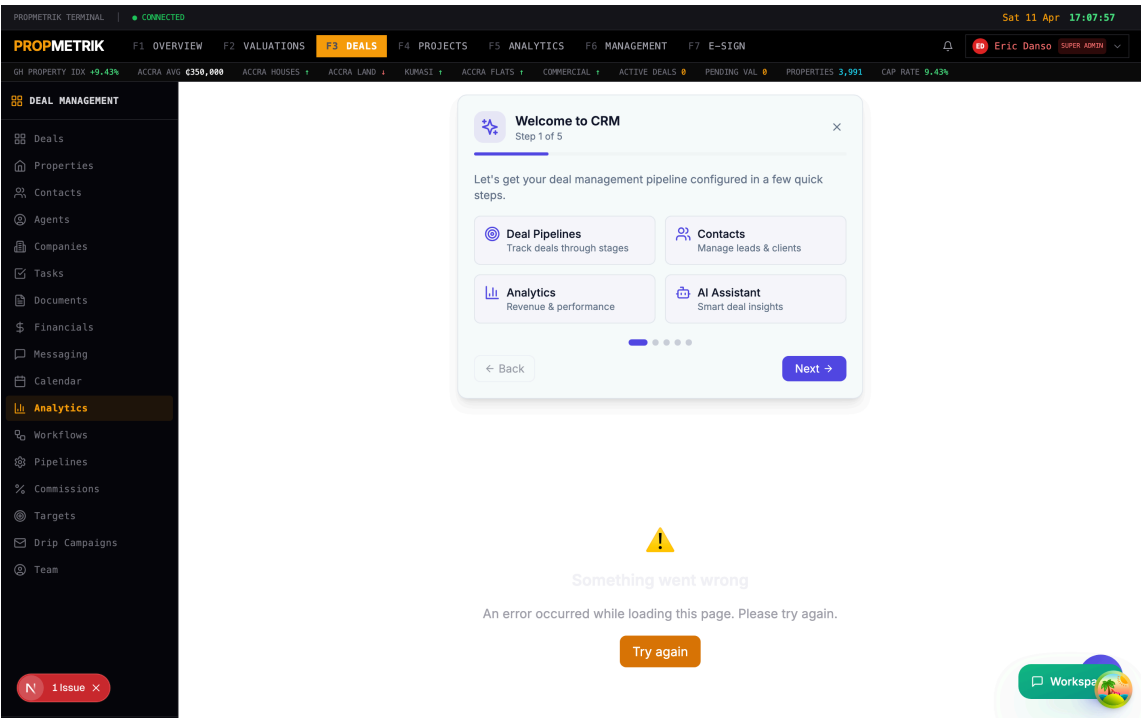
Individual valuer metrics are displayed in a ranked table:

- **Valuations completed** in the selected period
- **Average turnaround time** per valuer
- **Revenue generated** per valuer
- **Client satisfaction scores** (if feedback is collected)

Use the date range selector and filters to drill into specific time periods, property types, or geographic areas.

14.4 Deal Analytics

Track deal pipeline and CRM performance at **Dashboard > Analytics > Deals**.



Pipeline Overview

- **Total Pipeline Value** -- Sum of all active deals by stage.
- **Deal Count** -- Number of deals at each pipeline stage (lead, prospect, negotiation, under contract, closed).
- **Conversion Rates** -- Percentage of deals that move from one stage to the next.
- **Average Deal Size** -- Mean transaction value across closed deals.
- **Average Days to Close** -- Mean time from lead creation to deal closure.

Revenue Analytics

- **Closed Revenue** -- Total value of deals closed in the selected period.
- **Commission Revenue** -- Agent commissions earned.
- **Revenue by Agent** -- Individual agent performance ranking.
- **Revenue by Property Type** -- Which property categories generate the most deal volume.

Funnel Visualization

The deal funnel chart shows how many leads convert through each stage. Hover over any stage to see:

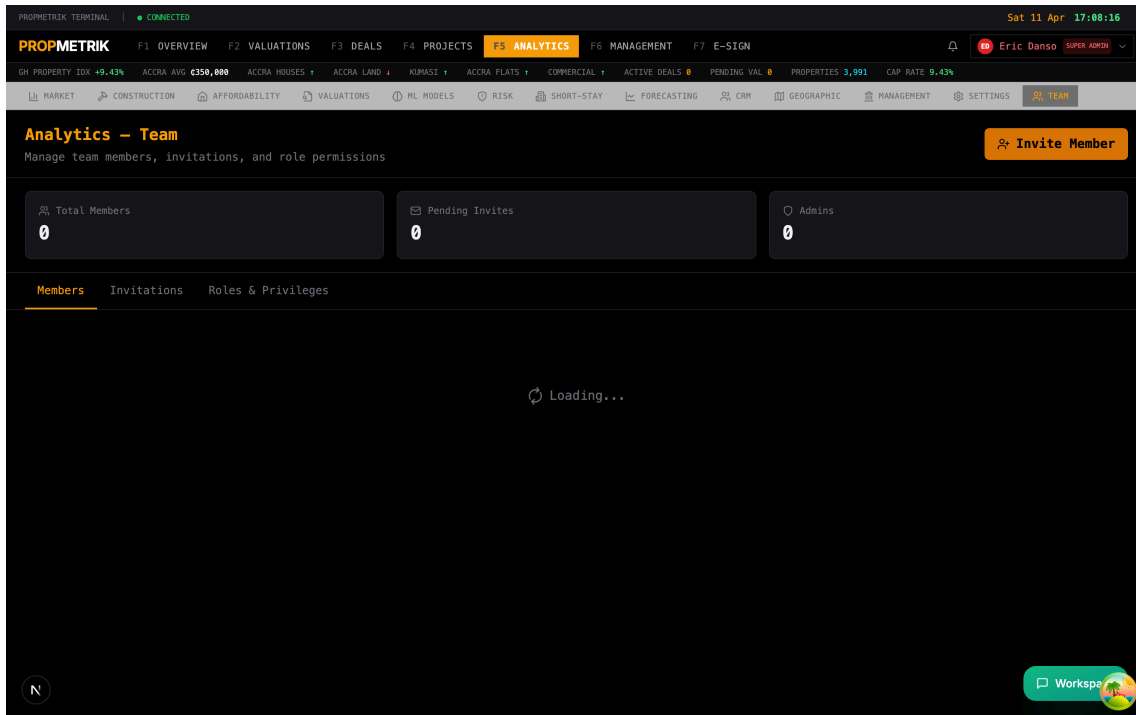
- Number of deals at that stage
- Total value
- Average time spent in stage

- Drop-off rate to the next stage

Tip: Monitor the drop-off rate between "Negotiation" and "Under Contract" stages closely. A high drop-off here often indicates pricing misalignment or documentation bottlenecks.

14.5 Team Analytics

Review team productivity and workload distribution at **Dashboard > Analytics > Team**.



Team Performance Metrics

- **Active Members** -- Number of team members currently working on projects, valuations, or deals.
- **Tasks Completed** -- Total tasks completed across all modules in the selected period.
- **Average Task Duration** -- Mean time to complete assigned tasks.
- **Overdue Task Rate** -- Percentage of tasks completed after their due date.

Individual Performance Cards

Each team member's card shows:

- **Module distribution** -- What percentage of their time is spent on projects, valuations, deals, or property management.
- **Task completion rate** -- Percentage of assigned tasks completed on time.
- **Active assignments** -- Current open items across all modules.
- **Utilization rate** -- Percentage of available capacity currently allocated.

Workload Heatmap

The heatmap visualization shows team capacity over time:

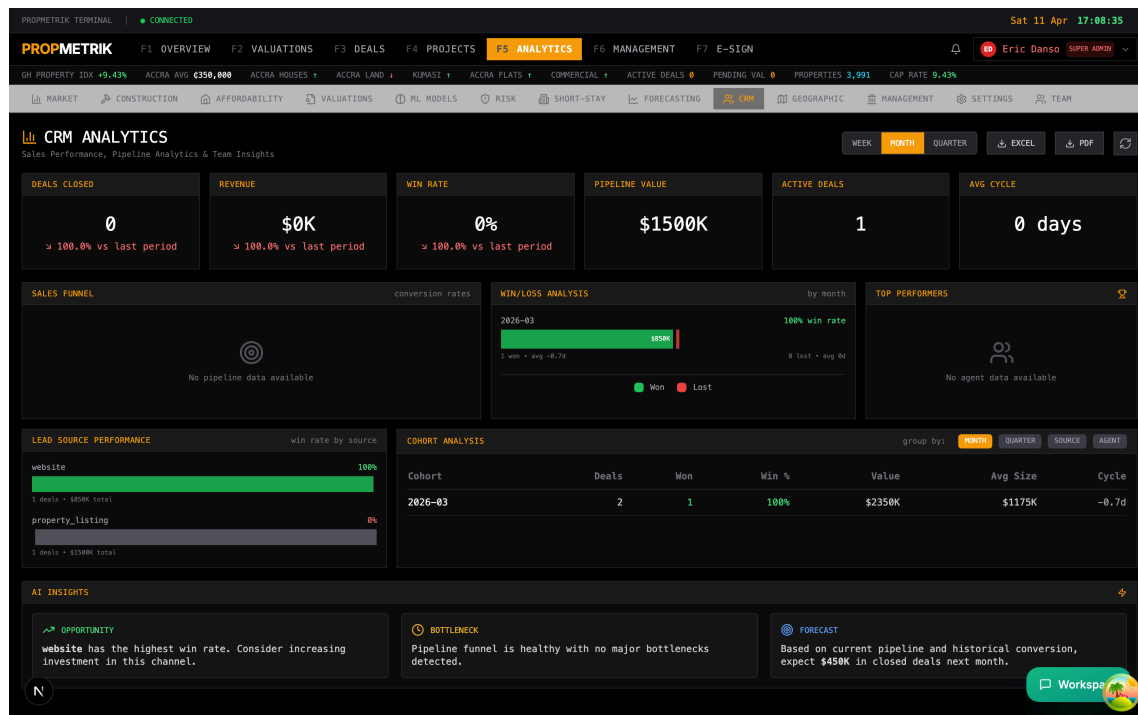
- **Green cells** indicate available capacity.

- **Amber cells** indicate near-full allocation.
- **Red cells** indicate over-allocation.

Use this view to identify team members who need workload redistribution and plan resource allocation for upcoming projects.

14.6 CRM Analytics

Deep-dive into customer relationship metrics at **Dashboard > Analytics > CRM**.



Contact Engagement

- **Total Contacts** -- Active contacts in the CRM database.
- **Engagement Score Distribution** -- How contacts are distributed across engagement tiers (hot, warm, cold).
- **Contact Growth** -- New contacts added over time.
- **Source Attribution** -- Where contacts come from (referral, web inquiry, walk-in, drip campaign, import).

Campaign Performance

If you are running drip campaigns (see Chapter 08 -- Deals & CRM):

- **Campaign Open Rates** -- Percentage of campaign emails opened.
- **Click-Through Rates** -- Percentage of recipients who clicked a link.
- **Conversion Rates** -- Percentage of campaign recipients who became active leads.
- **Revenue Attribution** -- Revenue from deals that originated from campaigns.

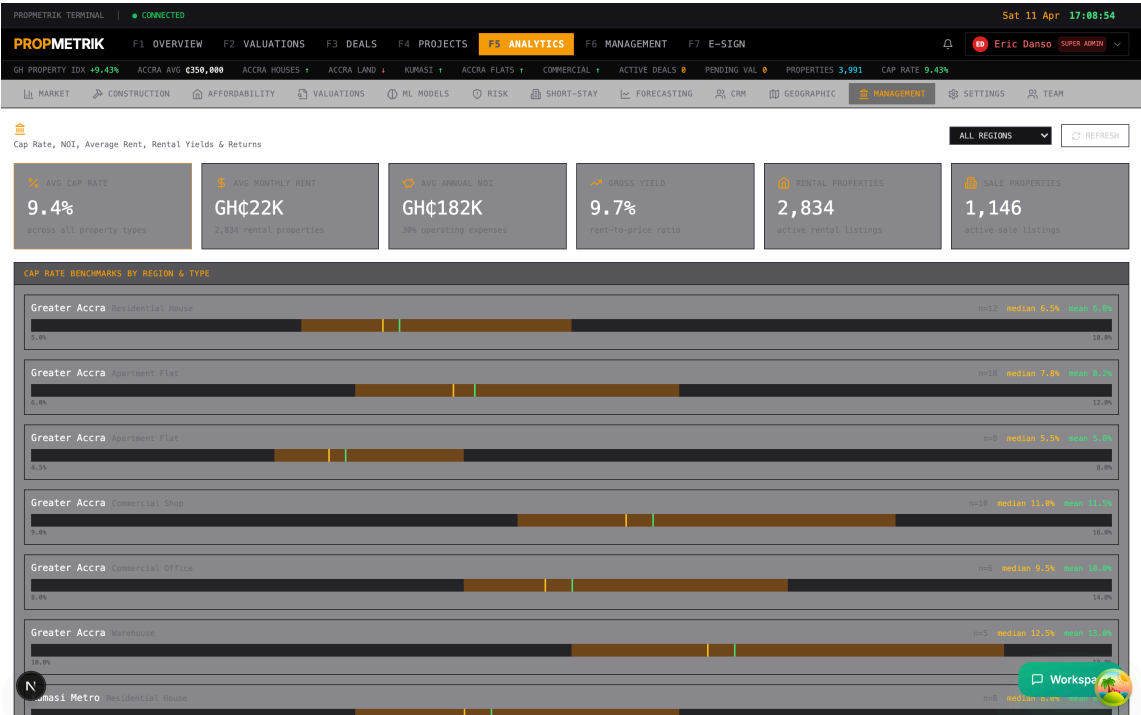
Lead Scoring

The CRM analytics page shows how the automated lead scoring model is performing:

- **Score Distribution** -- Histogram of lead scores across all active leads.
- **Conversion by Score Band** -- What percentage of leads in each score range convert to deals.
- **Top Scoring Factors** -- Which attributes most strongly predict conversion (property type interest, engagement frequency, budget range).

14.7 Management Analytics

Access executive-level dashboards at **Dashboard > Analytics > Management**.



Executive Dashboard

The management analytics page consolidates metrics from all modules into a single executive view:

- **Revenue Summary** -- Total revenue across valuations, property management fees, deal commissions, and project invoices.
- **Portfolio Performance** -- Occupancy rates, rental income, and NOI (Net Operating Income) across managed properties.
- **Project Health** -- Aggregate project status (on-track, delayed, at-risk) with budget variance summary.
- **User Growth** -- Platform adoption metrics including new users, active users, and user retention.

KPI Tracking

Define and track organization-level KPIs:

- Set **targets** for key metrics (e.g., "Complete 50 valuations per month" or "Maintain 95% occupancy").
- View **actual vs. target** performance with trend indicators.
- Configure **alerts** when KPIs fall below threshold.

Cross-Module Trends

The trend charts at the bottom show how key metrics evolve over time:

- Monthly revenue by module
- Valuation pipeline growth
- Property occupancy trends
- Deal conversion rate trends
- Team utilization patterns

Tip: Share the management analytics URL with your leadership team. The page updates in real-time and provides a comprehensive snapshot without requiring recipients to navigate multiple modules.

14.8 Valuation Leaderboard

Compare valuer performance at **Dashboard > Analytics > Valuations > Leaderboard**.

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PROPMETRIK

F1 OVERVIEW

F2 VALUATIONS

F3 DEALS

F4 PROJECTS

F5 ANALYTICS

F6 MANAGEMENT

F7 E-SIGN

GH PROPERTY IDX +9.43%

ACCRA AVG €350,000

ACCRA HOUSES +

ACCRA LAND +

KUMASI +

ACCRA FLATS +

COMMERCIAL +

ACTIVE DEALS 0

PENDING VAL 0

PROPERTIES 3,991

CAP RATE 9.43%

MARKET

CONSTRUCTION

AFFORDABILITY

VALUATIONS

ML MODELS

RISK

SHORT-STAY

FORECASTING

CRM

GEOGRAPHIC

MANAGEMENT

SETTINGS

TEAM

Eric Danso

SUPER ADMIN

SOMETHING WENT WRONG

An error occurred loading analytics. Please try again.

RETRY

DASHBOARD

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Leaderboard Rankings

The leaderboard ranks valuers by a composite performance score calculated from:

Factor	Weight	Description
Volume	30%	Number of valuations completed
Turnaround	25%	Average time from instruction to report delivery
Accuracy	20%	Deviation between estimated and actual sale prices (where data is available)

Revenue	15%	Total fee revenue generated
Client Feedback	10%	Average satisfaction rating from clients

Leaderboard Features

- **Time Period Selector** -- View rankings for the current month, quarter, year, or a custom date range.
- **Filter by Property Type** -- See rankings for specific property categories (residential, commercial, land).
- **Filter by Region** -- Compare performance within specific geographic areas.
- **Trend Indicators** -- Arrows show whether each valuer's rank is improving or declining compared to the previous period.

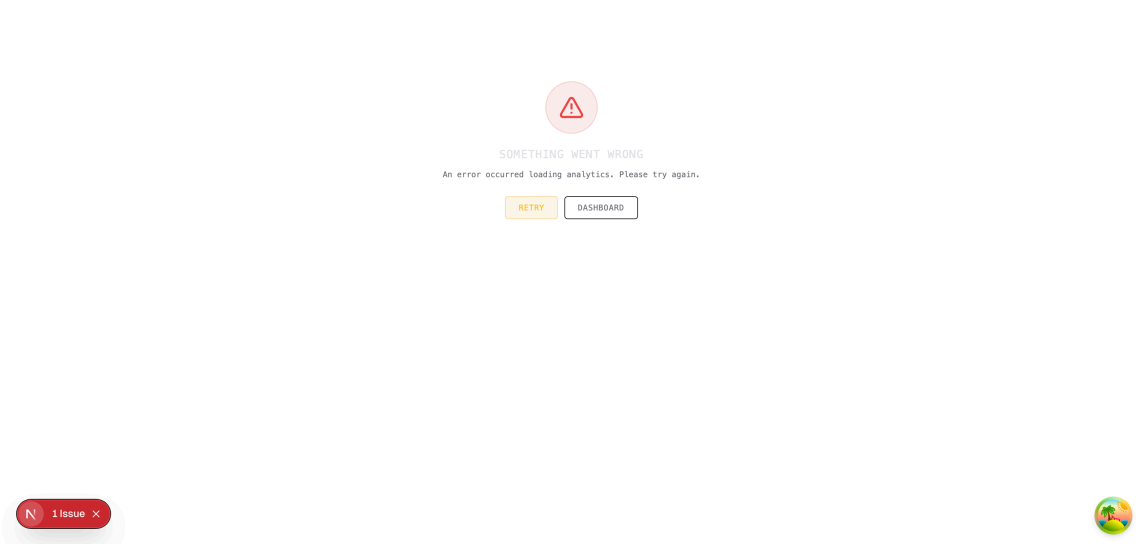
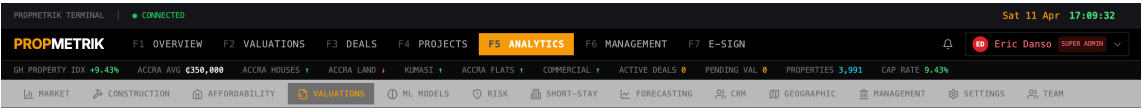
Individual Valuer Drill-Down

Click any valuer's name to see their detailed performance profile:

- Valuation history with property types and values
- Turnaround time trend chart
- Revenue contribution over time
- Active assignments and pipeline

14.9 Sensitivity Analysis

Model the impact of variable changes on property valuations at **Dashboard > Analytics > Valuations > Sensitivity**.



What Is Sensitivity Analysis?

Sensitivity analysis lets you test how changes to key assumptions affect a property's estimated value. This is essential for:

- Understanding which variables have the greatest impact on valuation
- Presenting best-case and worst-case scenarios to clients
- Stress-testing valuations against market shifts
- Validating model robustness

Running a Sensitivity Analysis

1. Navigate to **Analytics > Valuations > Sensitivity**.
2. Select the **valuation** to analyze (or enter property parameters manually).
3. Choose the **variables** to test:

Variable	Example Range
Capitalization Rate	+/- 1-3 percentage points
Discount Rate	+/- 1-3 percentage points
Rental Growth Rate	+/- 2-5 percentage points
Vacancy Rate	+/- 5-15 percentage points
Operating Expense Ratio	+/- 5-10 percentage points
Terminal Cap Rate	+/- 1-2 percentage points
Construction Cost Escalation	+/- 5-15%

4. Set the **step size** (how much each variable changes per increment) and the **range** (minimum and maximum values to test).
5. Click **Run Analysis**.

Interpreting Results

The output includes:

- **Tornado Chart** -- Horizontal bar chart showing which variables have the greatest impact on value. The widest bars indicate the most sensitive variables.
- **Data Table** -- Tabular output showing the estimated value at each variable level.
- **Spider Chart** -- Multi-axis chart showing how value changes as each variable moves from its base case.
- **Scenario Matrix** -- Two-variable grid showing values at intersections of two selected variables (e.g., cap rate vs. vacancy rate).

Exporting Sensitivity Analysis

- **PDF Report** -- Formatted report with charts, tables, and methodology notes suitable for client presentation.
- **Excel** -- Full data export with all calculated scenarios for further analysis.
- **Embed in Valuation Report** -- Insert the sensitivity analysis as an appendix to the main valuation report.

Tip: Always include a sensitivity analysis when presenting valuations to institutional clients or lenders. Focus on the 2-3 variables that show the greatest impact, and explain what market conditions could cause those variables to shift.

14.10 Additional Analytics Modules

PropMetrik includes several specialized analytics modules accessible from the main Analytics sidebar:

Affordability Analysis (Analytics > Affordability)

- Calculates the Housing Affordability Index (HAI) for different income brackets and locations.
- Shows monthly mortgage payment as a percentage of median household income.
- Compares affordability across regions, property types, and time periods.
- Powered by Bank of Ghana interest rate data and PropMetrik's property price database.

Construction Analytics (Analytics > Construction)

- Tracks the Construction Cost Index (CCI) over time.
- Breaks down cost drivers: materials, labor, fuel, equipment, and overheads.
- Compares construction costs across building types and regions.
- Provides cost forecasting based on historical trends and leading indicators.

Risk Assessment (Analytics > Risk)

- Property-level risk scoring incorporating:
 - **Flood risk** -- Based on NADMO incident data and geographic elevation.
 - **Market risk** -- Price volatility and liquidity measures.
 - **Credit risk** -- Tenant payment history and financial stability.
 - **Regulatory risk** -- Zoning compliance and permitting status.
- Portfolio-level risk aggregation and diversification analysis.
- Risk-adjusted return calculations for investment properties.

Geographic Analytics (Analytics > Geographic)

- Interactive map-based visualizations of property data.
- Heatmaps for property values, rental rates, transaction volumes, and risk scores.
- Neighborhood-level market statistics and comparables.
- Distance-based analysis for amenities, transport links, and infrastructure.

Summary

Task	Where to Go
Generate a project report	Projects > [Project] > Reports
Schedule automated reports	Projects > [Project] > Reports > Schedule
View PM compliance reports	Analytics > PM Reports
Analyze valuation production	Analytics > Valuations
Track deal pipeline	Analytics > Deals
Review team workload	Analytics > Team
Monitor CRM engagement	Analytics > CRM
Access executive dashboards	Analytics > Management

Compare valuer performance	Analytics > Valuations > Leaderboard
Run sensitivity analysis	Analytics > Valuations > Sensitivity
Check housing affordability	Analytics > Affordability
Track construction costs	Analytics > Construction
Assess property risk	Analytics > Risk
View geographic analytics	Analytics > Geographic